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SURVEY PAPER

Smart quadruped robotics: a systematic review of design, control, sensing and perception

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ABSTRACT

Legged robots in general, and quadrupedal structures in particular, have demonstrated high potential as a means of locomotion, possessing the ability to carry out tasks that traditional vehicles are unable to accomplish. In the last thirty years, legged locomotion technology has been advanced globally, leading to the creation of numerous equipment and techniques. This systematic review explores research studies from different academic databases, i.e. Web of Science, Scopus, and IEEE, to provide an overview of recent innovations and future directions in quadruped robotics. Advanced sensory systems, optimal structural design and smart control mechanisms are the key elements for enabling autonomous navigation and agile interaction of these robots with their environment. This systematic review explores the interconnected domains of structural design, control systems, and sensory integration. The analysis aims to identify common research themes and emerging trends by employing keyword co-occurrence and machine learning clustering techniques, providing a detailed understanding of the research landscape.

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Quadruped robotics; bio-inspired design; autonomous systems; model predictive control; reinforcement learning

List of acronyms

Acronym	Definition
AI	Artificial intelligence
ATRON	Advanced telerobotic operations network
BPNN	Back propagation neural network
COBOTS	Collaborative robots
COG	Center of gravity
CP	Capture point
CPG	Central pattern generator
DE	Deposition efficiency
DDP	Differential dynamic programming
DDPG	Deep deterministic policy gradient
DOC	Differentiable optimal control
DOF	Degree of freedom
DQN	Deep Q-Networks
DRL	Deep reinforcement learning
FEA	Finite element analysis
FL	Feedback linearization
FLCs	Fuzzy logic controllers
GPS	Global positioning system
HiLQR	Hybrid iterative linear quadratic regulator
HRI	Human-robot interaction
IMU	Inertial measurement unit
LIDAR	Light detection and ranging
LSM	Liquid state machines
LVIO	Lidar vision inertial odometer
M-TRAN	Modular transformable robotic system
MARL	Multi-agent reinforcement learning
MDR	Motion distortion removal
MIMO	Multiple-input multiple-output
ML	Machine learning
MPC	Model predictive control
NTSMC	Nonsingular terminal sliding mode control
NMPC	Nonlinear model predictive control
PCP	Pre-charged pneumatic
PGRL	Policy gradient reinforcement learning
PIBB	Probability-based black-box optimization
PID	Proportional–integral–derivative
PMF	Point-map fusion
PPO	Proximal policy optimization
QR	Quadruped robots
RL	Reinforcement learning
RBF	Radial basis function
SAC	Soft actor-critic
SLAM	Simultaneous localization and mapping
SISO	Single-input single-output
SR-CR	Soft rod-climbing robot
SRB	Single rigid body
TRPO	Trust region policy optimization
VQ-PMC	Vector-quantized primitive motor control

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WBC	Whole-body control
ZMP	Zero-moment point

Introduction

The fascination with the agile dynamics of living animals has long motivated robotics research, with the goal of replicating such abilities in robotic systems [1]. Recent advancements in robotics have led to the development of quadruped robots (QRs), which combine the stability of stationary platforms with the high mobility of dynamic locomotion systems. These robotic systems have already found significant applications ranging from industrial automation to search-and-rescue missions. However, realizing the full potential of these systems requires addressing the inherent challenges in their structural design, sensing, control, and locomotion systems [2,3].

QR structural designs range from articulated to compliant to hybrid, with recent trends emphasizing lightweight materials, modular components, and bio-inspired mechanisms. Advanced control strategies such as inverse kinematics and dynamics, zero-moment point (ZMP) control, model predictive control (MPC), and reinforcement learning (RL) are utilized to achieve real-time coordination of leg movements, balance management, and speed modulation [4].

Efforts in quadruped robotics focus on achieving locomotion that rivals that of natural quadrupeds. This objective necessitates an integrated approach encompassing actuation, sensing, material selection, and mechanical configurations that emulate biological motion. Stability and response to various perturbations are analyzed through detailed mathematical modeling and simulation, considering different gaits and terrains. Advanced control strategies facilitate the coordination of leg movements, balance management, and speed modulation in real-time [4]. Optimizing gait patterns, torque control, and shock absorption is imperative, alongside considerations for size, weight, speed, and energy efficiency. The transition of quadrupeds from controlled laboratory settings to unpredictable real-world environments underscores the importance of thorough performance validation [5,6]. Quadruped robotics, therefore, represents a synthesis of biomechanics, mechatronics, dynamics, and control theory, positioning it at the forefront of contemporary robotics research [7–10]. This review delineates the challenges and innovations that shape the development of dynamically dexterous QRs.

Systematic examination of the current state of quadruped robotics research forms the core of this review. Bio-mimetic design principles, inspired by nature, have been instrumental in endowing QR with versatility and adaptability akin to their natural counterparts [10]. Mathematical modeling and simulation contribute to an in-depth understanding of stability, gait transitions, and perturbation responses. The evolution of autonomous capabilities is furthered by advances in control strategies, including model predictive control, reinforcement learning, and sensor fusion. Hardware validations emerge as a pivotal aspect, marking the progression of quadrupeds into real-world applications.

Despite these advancements, quadruped robotics faces ongoing challenges. Achieving robust dynamic balancing and shock absorption while maintaining energy efficiency remains crucial [11]. The development of an integrated mechatronic design is necessary to balance factors such as weight, payload capacity, speed, and operational duration. Furthermore, enhancing sensory capabilities to interact seamlessly with unstructured environments is a significant area of focus. This review offers a comprehensive view, highlighting the promising directions to address these challenges. This review is unique in its comprehensive focus on the aspects of design, control, and sensing, providing a broader spectrum of analysis compared to other reviews that typically focus on a specific aspect [12,13]. Additionally, the use of systematic review methodologies, such as keyword co-occurrences and clustering, further distinguishes this study by offering a more nuanced and interconnected understanding of the field.

The structure of this paper is guided by the findings of the systematic review, ensuring a thorough examination of the field. Following a systematic analysis, we provide an overview of the design aspects of QRs. We then present a section discussing both traditional and advanced strategies in the control mechanisms of QRs. This is followed by a section on sensing and perception, which covers the latest advancements in sensor technologies and fusion techniques. Finally, the paper concludes by identifying the challenges and limitations faced in the development of quadruped robotics and proposes future directions and potential applications.

Methodology for the systematic review

To systematically investigate the current state-of-the-art in quadruped robotics, with a particular focus on design, control mechanisms, sensing and emerging trends, a comprehensive literature review process is adopted. This process employs text mining and systematic analysis methods akin to those delineated by [14–16].

Presented below is a table that describes the structured methodology employed in this systematic review (Table 1). The methodology encompasses a series of steps, commencing with the formulation of research

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Table 1. Key steps and descriptions of the systematic review process.

Step	Step title	Description and key elements
1	**ResearchQuestions**	To steer the direction of our review, key research questions are formulated. These questions aim to unravel significant aspects of quadruped robotics: • What constitute the current trends and developments in the design of QR? • In what ways have control mechanisms for QR evolved in recent years?Which emerging technologies and trends are currently influencing the future trajectory of quadruped robotics?
2	**Identificationof Keywords**	We establish a search query using Boolean operators to encapsulate the scope of our study: ('Quadruped Robot' OR 'Four-legged Robot' OR 'Robotic Dog' OR 'Canine Robot') AND ('Design' OR 'Engineering' OR 'Mechatronics' OR 'Kinematics' OR 'Dynamics') AND ('Control Systems' OR 'AI' OR 'Machine Learning' OR 'Autonomous Navigation' OR 'Sensor Integration')
3	**Selection ofResources**	The literature search encompasses widely recognized academic databases, known for their extensive coverage in the fields of robotics and technology: • Web of Science (https://www.webofknowledge.com/) • Scopus (https://www.scopus.com/) • IEEE (https://ieeexplore.ieee.org/)
4	**Timeframe**	The scope of the literature review is confined to publications spanning the period from the year 2000 through 2023.
5	**Inclusion andExclusion Criteria**	**Inclusion Criteria:** • Studies specifically focusing on the design and control of QR. • Papers discussing emerging trends and technologies in quadruped robotics. • Publications from the timeline 2000 to 2023. **Exclusion Criteria:** • Studies not specifically addressing QR. • Papers that focus solely on other types of robotics without direct relevance to quadruped robotics. • Non-English publications.
6	**Article Selectionand Analysis**	The initial search based on the keywords will yield a broad range of publications. We will apply the exclusion criteria to this initial set, followed by a thorough review of the remaining articles against our inclusion criteria. The final selection of papers will be analyzed for quality and relevance.
7	**Co-Occurrence andClustering Analysis**	Following the selection of relevant papers, we will conduct a co-occurrence and clustering analysis to identify prevalent themes and trends within the field. This analysis will help in understanding the concentration of research efforts and emerging areas within quadruped robotics.

questions and culminating in a co-occurrence and clustering analysis.

We conducted searches in the Web of Science, Scopus, and IEEE databases using Boolean search queries that covered key quadruped robotics terms from 2000 to 2023. Inclusion and exclusion criteria were focused on QRs-related papers.

In the subsequent section, the outcomes of the systematic literature review are presented.

Systematic review analysis

Following the execution of a literature search using the previously defined search query in the three databases, the findings revealed both common trends and notable differences across the database sources. The initial years (2000–2018) exhibited a relatively modest number of publications in all databases, reflecting the nascent stage of quadruped robotics research (Figure 1). Over time (2018–2023), there was a marked increase in the number of publications, indicating a growing interest and significant advancements in the domain.

By analysis of word clouds from the Web of Science, Scopus, and IEEE datasets, it was observed that the central term is 'Legged locomotion,' indicating a strong focus on the strategic aspect of quadruped control and navigation. The other core terms are 'control system,' 'robot kinematics,' 'sensing systems,' 'Reinforcement learning,' and 'machine learning,' each sizable and suggesting major themes in the field of quadruped research. This emphasizes the importance of machine learning, control systems, and design processes, as shown in Figure 2.

The co-occurrence analysis from the selected Scopus articles, depicted in Figure 3(a), offers a nuanced exploration into quadruped robotics, focusing on design, control, and locomotion. We considered the Scopus database since it contains the highest number of papers. The emphasis on 'QR' and 'Multipurpose Robots' reflects a concerted effort to enhance their structural design for varied applications, from industrial to search and rescue tasks. The significance of 'Controllers' reveals the critical role of control systems for optimal performance across diverse environments. Furthermore, the analysis of 'Locomotion' and 'Legged Locomotion' illustrates the endeavor to replicate the efficient and adaptable movement patterns of natural quadrupeds, aiming for operational excellence and potential enhancements in robotic mobility. This involved blend of design innovation, control precision, and locomotive agility encapsulates the quest for robots that mimic the capabilities of their biological counterparts. In Figure 3(b), a systematic quantification of average normalized citations is observed, with 'sensor fusion' emerging as the preeminent topic within quadruped robotics literature in Scopus database, as indicated by its leading citation metric. Subsequently, terms such as 'robot sensing system' and 'optimization

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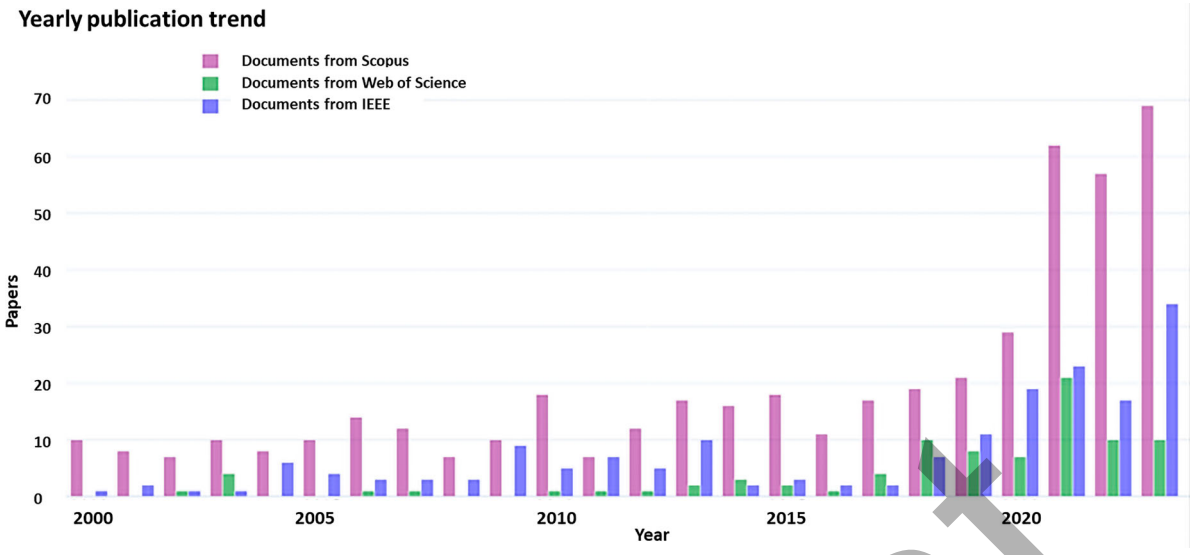


Figure 1. Yearly publication trends from Web of Science, Scopus, and IEEE databases.

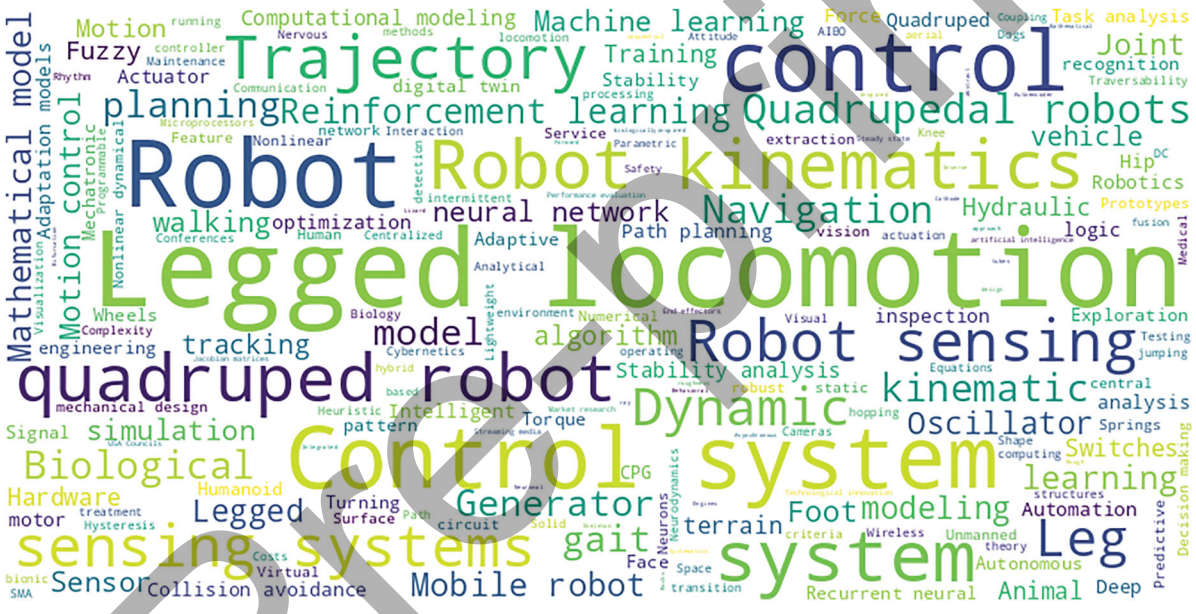


Figure 2. Word cloud visualization of the literature search from the databases.

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and optimal control' are acknowledged for their significant scholarly impact, denoted by their respective citation averages. The graph clearly favors the domains of sensory integration, control optimization, and movement efficiency within robotic research.

The clustering analysis conducted on the selected papers reveals insightful patterns and relationships within the domain of quadruped robotics. As illustrated in Figure 4(a), the clustering analysis conducted in Web of Science identifies five primary research domains within the collected papers, each denoted by a unique color (purple, blue, yellow, green, and red). The spatial arrangement of terms within this analysis underscores their interconnection, reflecting the methodological foundation previously outlined. Each cluster's thematic focus is discerned through the coherence of associated keywords, offering insights into prevailing research trends and areas of concentrated investigation in the field. The reorganization of research clusters into thematic pairs offered a nuanced perspective on quadruped robotics, encapsulating the progression from foundational dynamics and modeling to practical locomotion and control. The first pairing, 'Foundations and Adaptations,' elucidates the essential principles and their applications in dynamic environments. The second, 'Design and Mechanics for Locomotion,' underscores the interplay between design considerations and mechanical algorithms essential for adaptive movement. Lastly, 'Stability,

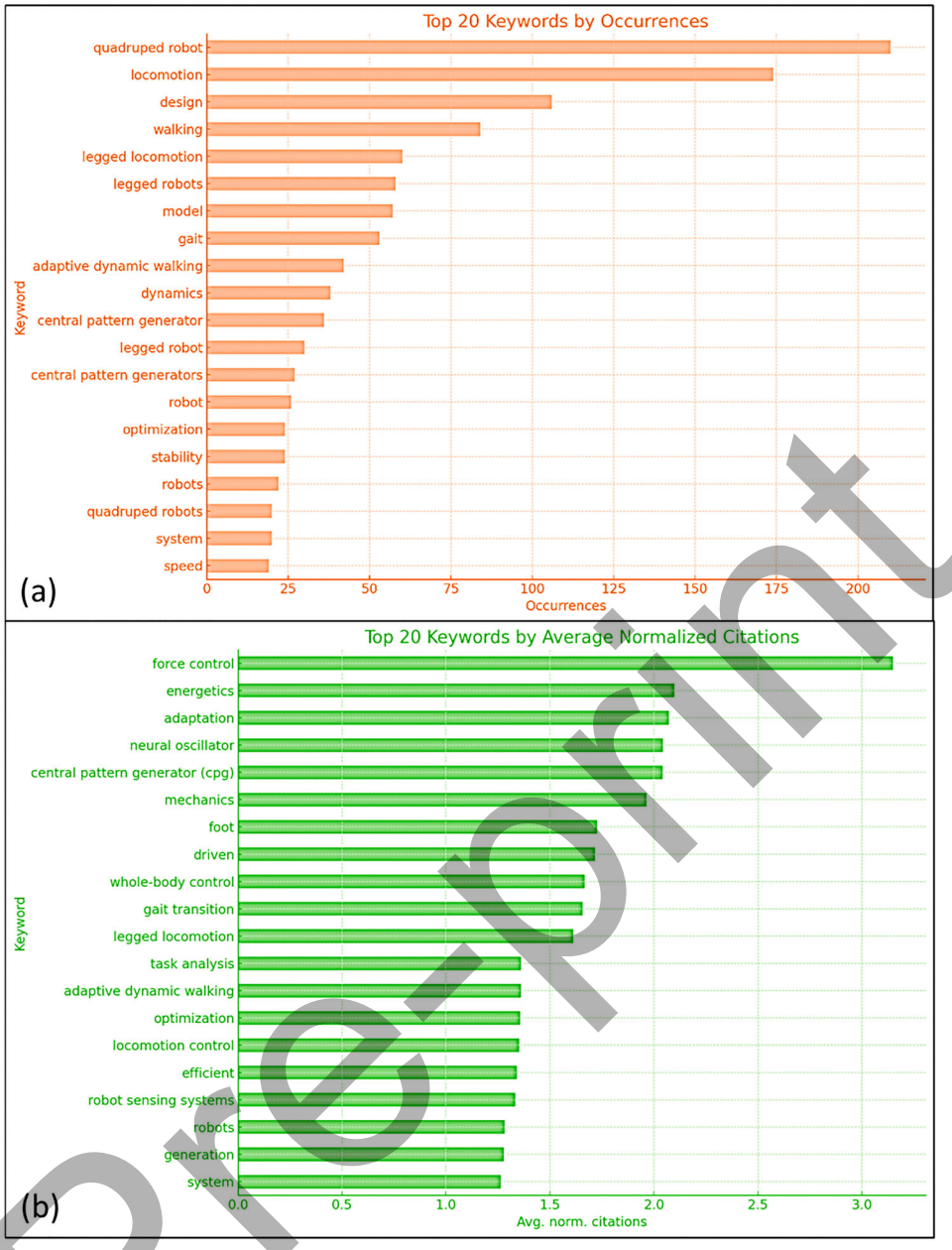


Figure 3. Co-occurrence analysis of the proposed article selection from Scopus database: (a) Occurrences, (b) Average Normalized Citations.

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Control, and Transition' highlights the critical aspects of maintaining stability and managing transitions, pivotal for the operational versatility of QR.

The analysis of top keywords, extracted from Scopus, across various clusters (Figure 4(b)) has led to the proposition of preliminary topics, each reflecting distinct research domains within quadruped robotics. These topics range from 'Bio-inspired Robotics and Locomotion,' highlighting the intersection of robotics with natural locomotive principles, to 'Quadruped Robot Design and Efficiency,' which encapsulates studies focused on actuation, energy efficiency, and design methodologies. Additionally, a unification of clusters into broader categories such as 'Robotic Locomotion and Analysis' suggests a thematic consolidation, emphasizing advanced locomotion techniques and the integration of neural analysis. This organization underscores the multifaceted nature of quadruped robotics research, spanning technical, biological, and efficiency-oriented inquiries.

Notably, the robot design and learning with locomotion and actuation cluster underscored the field's evolution towards a more integrated approach to robot mobility and cognitive capabilities. The interplay between 'quadruped robotics system and learning' and 'dynamics and kinematics of legged robots' highlighted a concentrated effort to enhance the efficiency and responsiveness

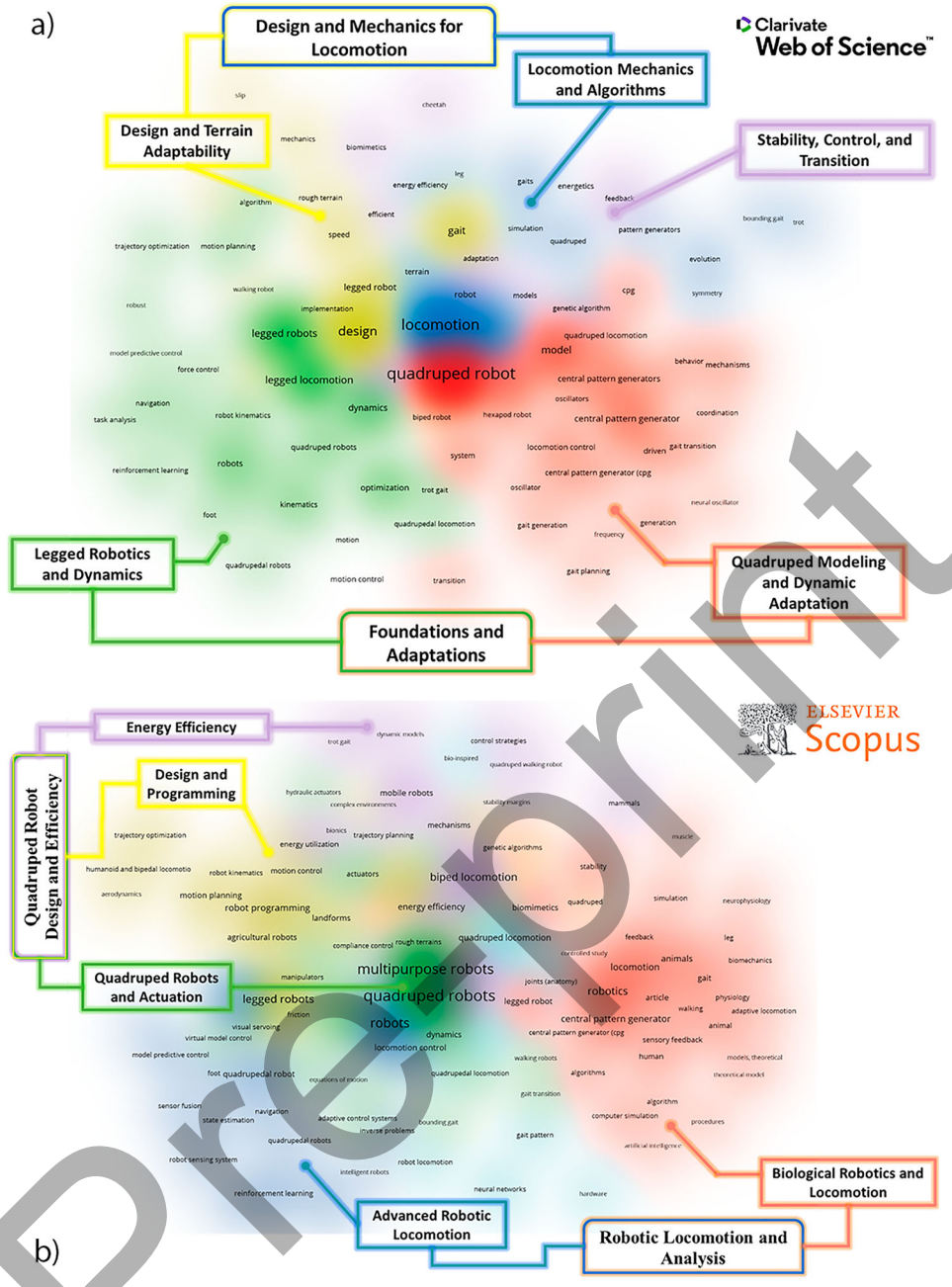


Figure 4. Clustering analysis of the collected article selection from (a) Web of Science, (b) Scopus, and (c) IEEE datasets.

of robotic systems. This convergence signifies a move towards a sophisticated equilibrium between computational control algorithms and the physical dynamics of robot movement, essential for navigating unpredictable terrains. The systematic review analysis, drawn from Web of Science, Scopus, and IEEE databases, has indicated three primary areas of focus for further investigation: the design aspects of QR, their control mechanisms, and sensing and perception capabilities. Subsequent sections will provide an in-depth examination of these critical topics, reflecting the current research trajectory in quadruped robotics. The structure of the subsequent sections is directly derived from the systematic review results, ensuring consistency and coherence throughout the paper.

Design aspects of quadruped robots

Mechanical and material design are vital for quadruped robotics' efficiency and functionality. A QR mechanism mirrors the biologic skeleton structure of quadrupedal animals by connecting four manipulators that form legs to a single component or link that forms the body [17,18].

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- ***Structural configuration***: Typically, QR integrate various link segments and joints in a legged structure (Figure 5). One of the key components of a QR mechanism is the leg topology. The leg topology, i.e. its structural assembly of joints and links, determines the configuration of the articulated legs in the base link. This configuration has a direct impact on the mobility factors that affect stability, flexibility, and other characteristics. Three primary categories of leg topologies [18] are the mammal-type and crawling-type topologies, as well as their combination crawling-mammal type. The design of joints and limbs is also pivotal in defining the robot's locomotion capabilities. This includes the optimization of limb length and joint range of motion for efficient gait patterns [19]. The incorporation of adaptive and compliant mechanisms, such as variable stiffness actuators, can enhance the robot's ability to navigate complex and unstructured terrains by adjusting rigidity dynamically.
- ***Actuation mechanisms:*** Actuators are the muscles of QR, and their selection is guided by factors such as power-to-weight ratio, response time, and energy efficiency [20]. Hydraulic [21], pneumatic [22], and electric actuators [23,24] are the most commonly used (Figure 6), each with their specific advantages and constraints. For instance, hydraulic actuators provide high force output but at the cost of increased system complexity and weight, whereas electric actuators offer precise control and simplicity but with limitations in force generation.
- ***Material selection:*** The choice of materials in QR is a critical aspect that influences not only the weight and durability but also the robot's interaction with its environment [25]. Advances in materials science have led to the utilization of lightweight composites, high-strength alloys, and smart materials. The application of bio-inspired materials that mimic the properties of natural muscles and bones, such as dielectric elastomers, has also gained traction. The selection is often a trade-off between mechanical properties (like strength and stiffness) (Figure 7), weight, and the robot's intended operational conditions (such as exposure to water, dust, or varying temperatures) [2].
- ***Integration of sensors:*** The mechanical design must also account for the integration of sensors, processors, and wiring without compromising the robot's mobility [26–29]. This includes strategic placement of sensors (Figure 8) for optimal environmental perception and designing conduits for wiring that do not hinder the robot's movements.

Comparative analysis of different structural designs in quadruped robots

Researchers have extensively studied QRs and explored various structural designs, each tailored to specific functions and environments. This comparative analysis examines the differences, advantages, and limitations of these designs. Added to the comparative analysis, an extensive list of works summarizing key research on QRs,

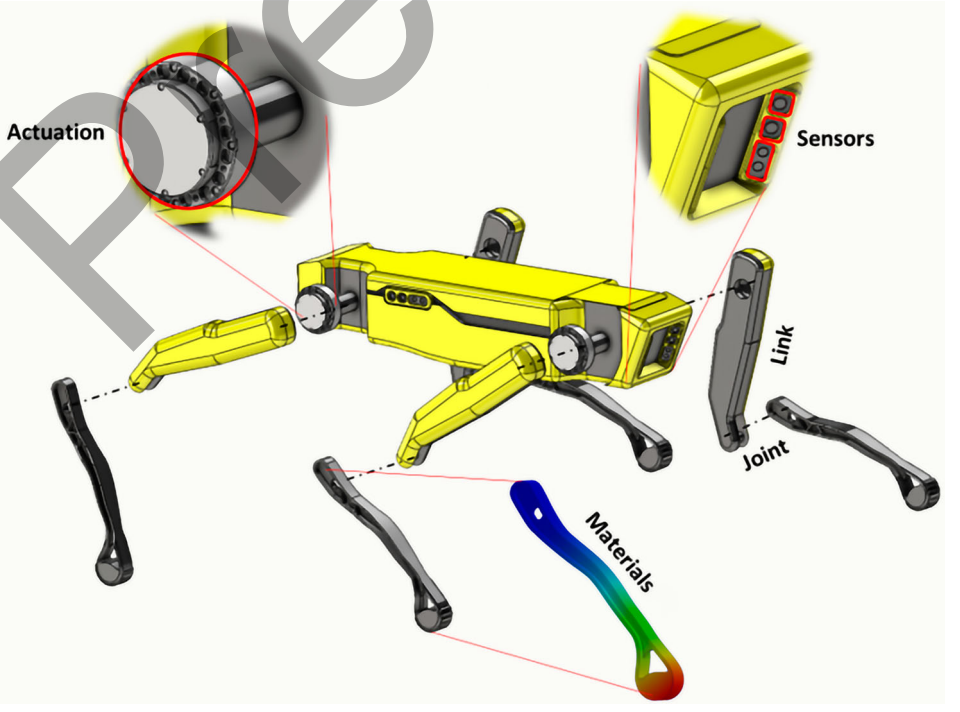


Figure 5. Schematic of different design aspects in a QR.

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Figure 6. Common types of actuators in robotics.

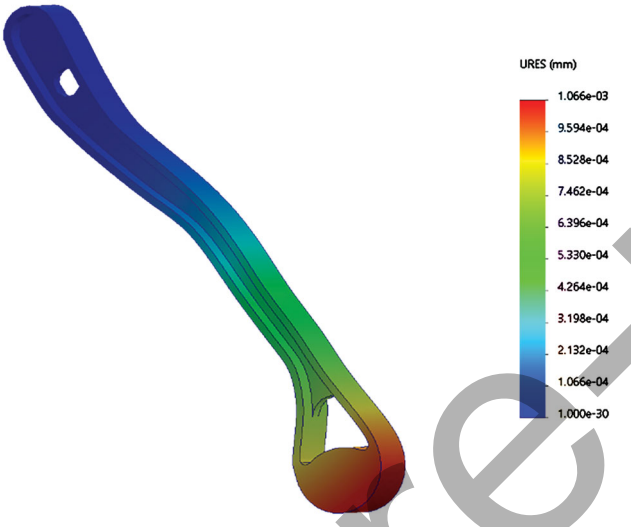


Figure 7. Example of stress distribution field of a QR leg.

including their focus, key findings, methodologies, and specific robot models used is provided in Table 2.

1. ***Bio-inspired designs:*** This category of robots is among the most prominent in the QR field as they replicate the anatomy and locomotion of biological quadrupeds. Canine-inspired robots, like Boston Dynamics' Spot [30], are designed for versatility and mobility across diverse terrains, featuring sturdy, segmented legs with multiple degrees of freedom (DOFs) and flexible leg orientations that provide stability and adaptability. In contrast, feline-inspired designs, such as the MIT Cheetah [31], emphasize speed and agility by incorporating spine-like structures that allow for rapid, efficient movement. These designs typically have fewer segments per limb but include an actuated spine that enables dynamic flexibility and enhanced propulsion. The main advantage of bio-inspired QR designs is their

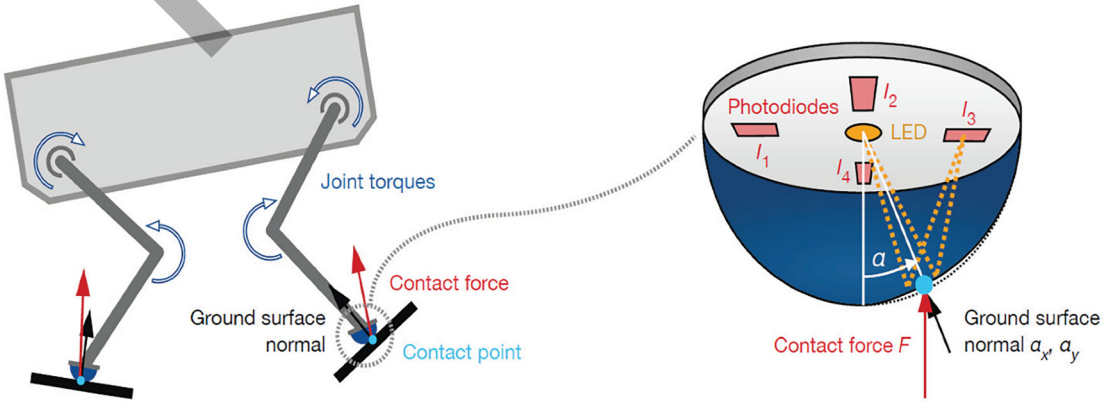


Figure 8. Optical sensor attached to the QR foot. This image was reproduced from [29] (open access).

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Table 2. Comparative analysis of QR structural designs and list of additional works.

Design category	Description	Advantages	Limitations/Challenges
Bio-inspired designs 	Mimics anatomy and locomotion of biological quadrupeds.	Biomimetic efficiency, natural maneuverability.	Complexity in replicating biological muscle/joint interactions.
Mechanically simplified designs 	Simplified joint structure, more rigid body.	Easier to manufacture and maintain, cost-effective.	Limited mobility and adaptability.
Modular and reconfigurable designs 	Flexible design with interchangeable parts.	Versatile, adaptable to various tasks.	Increased mechanical and control complexity.
Specialized function designs 	Tailored for specific tasks like heavy lifting or climbing.	Highly effective in designated roles.	Lack of general versatility.
Material-driven designs 	Utilizes novel materials for enhanced adaptability.	Can navigate tight spaces, adaptable to surfaces.	New materials may have durability or control complexities.

Other related works

Bio-inspired designs

Ref	Focus	Key findings	Methods	Robot model
[30]	Interaction between spinal CPG, sensory feedback, and supraspinal drive signals in quadruped locomotion	Direct supraspinal modulation enhances gap crossing success; CPG dynamics contribute to gait smoothness and energy efficiency; front feet sensing crucial for gap crossing	Deep reinforcement learning (DRL), neural network policy	QRs (simulation)
[31]	Learning capabilities and robustness of Liquid State Machines (LSMs) in robot-environment interactions	Developed an efficient robot state estimation method through reservoir computing; highlighted LSM's efficacy even under faulty sensory conditions	Reservoir computing, simulation results	Simulated QR, Lilibot model
[32]	Development of a bio-inspired quadruped locomotion controller for omnidirectional movement and adaptive path planning	Demonstrated effectiveness in generating desired trajectories and dynamic obstacle avoidance	Sensory-motor visual feedback, dynamical systems theory	AIBO platform, simulated QR
[33]	Enhancement of Differential Dynamic Programming (DDP) with multiple shooting techniques	Achieved local quadratic convergence, improved convergence performance, and facilitated dynamic jumping in QRs	Multiple shooting techniques, adaptive merit function, penalty method	QR, quadrotor, acrobot, manipulator
[34]	Nonlinear Model Predictive Control (NMPC) for specialized motion skills and standard locomotion in QRs	Enabled execution of multiple agile skills and transitions; enhanced agility and adaptability	Hybrid kinodynamic model, constrained Differential Dynamic Programming (DDP) solver	Unitree A1, MIT Mini Cheetah
[35]	Nonlinear Model Predictive Control (NMPC) for specialized motion skills and standard locomotion in QRs	Enabled execution of multiple agile skills and transitions; enhanced agility and adaptability	Hybrid kinodynamic model, constrained Differential Dynamic Programming (DDP) solver	Unitree A1, MIT Mini Cheetah
[36]	Overview of ANYmal, a QR designed for harsh environment operations	Showcased proficiency in diverse climbing maneuvers, walking gaits, dynamic trots, and jumps; highlighted durability, precise torque control, and low energy consumption	Rotating lidar sensors, compact force sensors, compliant joint modules with integrated electronics	ANYmal
[37]	Development of Lilibot, an open-source reconfigurable QR with multiple sensory feedback	Demonstrated autonomous adaptation to various locomotion challenges; highlighted potential as a research and educational platform for spontaneous locomotion, posture stabilization, and compliant behavior	Modular components for flexible leg orientation, real-time sensory feedback, adaptive neural controller	Lilibot

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Table 2. Continued.

Mechanically simplified designs				
[38]	Direct force control in QR locomotion using a straightforward linear policy	Developed a linear policy for generating end-foot trajectory parameters and centroidal wrench, validated in both simulation and real-world experiments	Linear function approximator for policy representation, quadratic program for force distribution, centroidal dynamics-based MPC, imitation learning	Stoch3
[39]	Open-source dynamic QR with high control bandwidth and accessible design	Constructed from off-the-shelf components and 3D-printed parts; confirmed robustness through FEA and continuous walking tests; supports affordable and customizable research	12 torque-controlled quasi direct drive joints, onboard depth sensor, GPU-equipped computer, finite element analysis (FEA)	PADWQ
[40]	Development of Stanford Pupper, a low-cost agile QR for benchmarking and education	Features torque-controllable brushless motors; built from off-the-shelf components and 3D-printed parts; introduced benchmarks for high-speed locomotion and robustness	Torque-controllable brushless motors, 3D printing, benchmarks for Sprint and Scramble tasks	Stanford Pupper
[41]	Design, development, and computational simulation of a 12DoF QR	Developed a low-cost, modular QR using 3D-printed ABS plastic parts; utilized a Watt six-bar linkage mechanism for enhanced walking motion; performed kinematic and dynamic performance analysis	3D-printed ABS plastic parts, Watt six-bar linkage mechanism, SolidWorks for structural design, PyBullet physics engine for simulation	Iron Dog Mini
[42]	Design and performance of MIT Super Mini Cheetah	Developed a small (sub-10 kg), low-cost (sub-10k$) quadrupedal robot for dynamic locomotion with precise force and impedance control.	Utilized commercial-off-the-shelf components and rapid prototyping methods.	MIT Super Mini Cheetah
Modular and reconfigurable designs				
[43]	Design of a reconfigurable multi-legged walking robot	MiniQuad I is a modular robot reconfigurable in both leg and body, enhancing adaptability, maintainability, and extensibility. Simulations and experiments validated its reconfigurability and functionality in different tasks.	Modular design, biomimetic control, simulations, and experiments	MiniQuad I
[44]	Design and locomotion capabilities of UBot	Developed a new reconfigurable modular robotic system (UBot) capable of multiple locomotion modes such as cross, loop, quadruped, and snake-type.	Physical experiments, universal joint-based cubic modules with two rotational joints and reliable automatic connecting mechanism.	UBot
[45]	Adaptive locomotion in self-reconfigurable modular robots	Distributed learning strategy achieved optimal gaits within 15 min for most configurations; morphology-independent.	Distributed reinforcement learning; experiments in simulation and physical robots.	ATRON, M-TRAN (simulated and physical)
Specialized function designs				
[46]	Small morphing leg-wheel hybrid QR for enhanced exploration capabilities	Enhanced mobility and manipulation capabilities; adaptable across different terrains with dual-arm cooperative manipulation	Morphing movement/manipulation platform, sensors, embedded control system, kinematic equations, ultrasonic sensors, WI-FI camera	Morphing leg-wheel hybrid QR
[47]	Advanced navigation capabilities for BigDog QR in rough terrain	Successfully navigated unstructured forest environments, perceiving and avoiding obstacles; reached target locations in 23 out of 26 attempts	Laser scanner, stereo vision system, perception and navigation algorithms, field tests	BigDog
[48]	An Untethered Soft Robotic Dog Standing and Fast Trotting with Jointless and Resilient Soft Legs	Developed an untethered soft robotic dog with bioinspired dynamic trotting gait using jointless, resilient soft legs; demonstrated stability and robustness over various terrains and high-impact resistance	Soft robotics, pre-charged pneumatic (PCP) actuation, dynamic trotting gait, impact resistance	Soft robotic dog (Figure 12)

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Table 2. Continued.

		Material-driven designs		
[49]	Hybrid robotic system integrating hard and soft subsystems	Demonstrated rapid flat terrain locomotion and object retrieval; showcased synergy of hard and soft robotic systems with ease of assembly and control	Wireless control, morphing movement/manipulation platform, sensors, kinematic equations	Hybrid robotic system (iRobot Create and soft quadruped)
[50]	Quadruped soft rod-climbing robot for specialized environments	Achieved vertical and horizontal climbing with speed and load-bearing capabilities; showcased adaptability in complex climbing tasks	Bending actuators, telescopic actuator, pneumatic clenching, flexible bellows, elastic telescopic layer, strain-limiting layer	SR-CR (Soft Rod-Climbing Robot)
[51]	Distributed and morphology-independent strategy for adaptive locomotion in self-reconfigurable modular robots	Developed a distributed reinforcement learning strategy that enables each module to independently optimize its actions based on local information and a shared global reward signal. The strategy was effective in both simulated and physical robots, demonstrating adaptability to various configurations and module faults.	Distributed reinforcement learning, simulation, and physical experiments	ATRON and M-TRAN robots

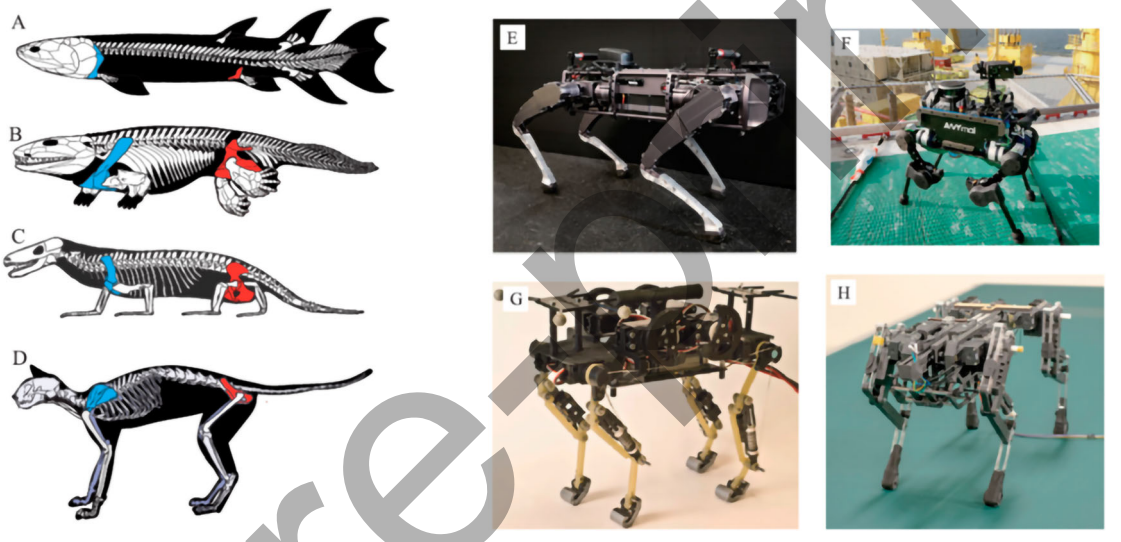


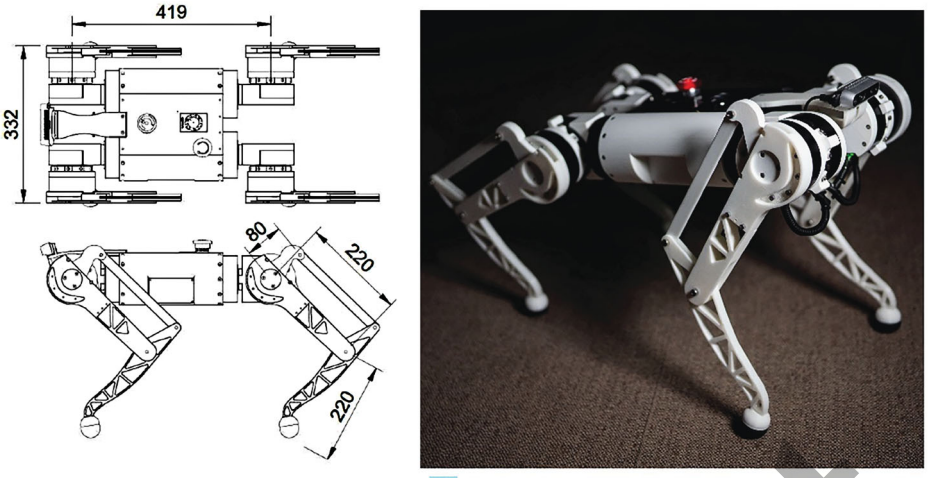
Figure 9. Trunk, forelimb, and hindlimb connectivity in animals and robots: (A) prehistoric Eusthenopteron, (B) tetrapod Ichthyostega, (C) Primitive Procynosuchus delaharpeae, and (D) domestic cat Felis catus. The red area represents the shoulder girdle, and the red area represents the pelvis. (E) Vision 60 v3.9 by Ghost Robotics, (F) ANYmal by ANYbotics, and (G) Cheetah-cub feature rigid connections between the trunk and limbs. (H) A QR with a flexible shoulder connection similar to a cat. This image was reproduced from [2] (open access).

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biomimetic efficiency, which enables them to negotiate challenging surroundings in a way similar to their natural biological counterparts. However, accurately reproducing the complex muscle/joint interactions seen in living entities remains challenging.

A. Fukuhara et al. [2] studied the differences and similarities in the anatomical topologies of QRs and animals. The review emphasized that robots commonly use modular designs with single-functional modules, whereas animals use situation-dependent multifunctionality with body parts capable of performing multiple roles. The study examined the anatomical features of limbs and spines, highlighting how animal adaptations, such as the flexible shoulders of mammals and the biarticular muscles, played a role in their highly adaptable behaviors (Figure 9). The review proposed that incorporating a 'polysemantic body' design in robotics that mimics the multifunctional characteristics of animal anatomy could highly improve the adaptability and efficiency of robots in various environments.

In a review paper [52], Y. Li et al. analyzed the development and challenges of mammal-inspired QRs. It traced the timeline of notable QR models, including Boston Dynamics' BigDog and MIT's Cheetah, underscoring the benefits of utilizing hydraulic actuators in



(a) Overall Dimension (b) The assembled robot

Figure 10. The PADWQ robot mechanical design. This image was reproduced from [39] (open access).

payload capacity and dynamic performance. Among the key trends are bio-mimetic design, lightweight high-power actuators, high mobility, intelligence, and human-robot interaction. Technical challenges that were addressed included environmental perception, gait planning, dynamic locomotion control, mechanical design, and integrated hydraulic drives. The review concluded by highlighting the need for innovations in actuator systems and dynamic stability to improve the efficiency and adaptability of QRs.

2. ***Mechanically simplified designs:*** These designs prioritize simplicity and cost-efficiency rather than biomimetic complexity. They frequently exhibit sturdier body blocks and a simplified joint configuration, making them more convenient for building and maintaining. Educational and research environments commonly use these QRs. Although these robots do not possess the advanced mobility of their bio-inspired counterparts, they provide a reliable and easily accessible platform for conducting fundamental studies on QR locomotion.

In [53], J. M. Morrey et al. presented 'Mini-Whegs,' which are small QR robots that draw inspiration from the locomotion principles of cockroaches. These robots utilized three-spoked 'whegs,' which combined the velocity of wheels with the climbing proficiency of legs. Mini-Whegs achieved high-speed locomotion (more than 10 times their body length per second), successfully scaled tall obstacles, and functioned in various positions. A modified version of Jumping Mini-Whegs included a mechanism for overcoming larger obstacles. They featured a single propulsion motor, a rectangular chassis, and durable whegs. They successfully traversed diverse landscapes, endured impacts, and transported loads that were twice as heavy as their weight. Future enhancements will focus on improving traction on slippery surfaces, making the chassis waterproof, and developing autonomous systems.

Another good example of simplified design is illustrated by the work of J. Kim et al. in [39]. They developed an open-source dynamic QR, PADWQ (Figure 10), which is characterized by its 12 torque-controlled quasi-direct drive joints with high control bandwidth. Unlike other dynamic QRs that rely on custom actuators and machined metal structural parts, PADWQ is constructed entirely from off-the-shelf components and standard 3D printed plastic structural parts. This design choice significantly enhances the robot's accessibility, allowing for rapid distribution and duplication without requiring advanced machining processes. To ensure the durability of the plastic structural components under the demands of dynamic locomotion, finite element analysis (FEA) was applied to the leg structural parts, complemented by a continuous walking test on the physical robot. Both evaluations confirmed the robot's robustness. This development is aimed at supporting a broad spectrum of researchers and engineers in need of an affordable, highly capable, and easily customizable QR.

3. ***Modular and reconfigurable designs:*** Modular QRs emphasize adaptability and flexibility. These robots have the ability to modify their compositional form in order to adapt to a variety of tasks and environments. Their modular designs allow for the addition, removal, or rearrangement of limbs and other components, thereby enabling a wide range of applications. While modularity offers notable advantages, it often increases mechanical complexity and the degree of control required.

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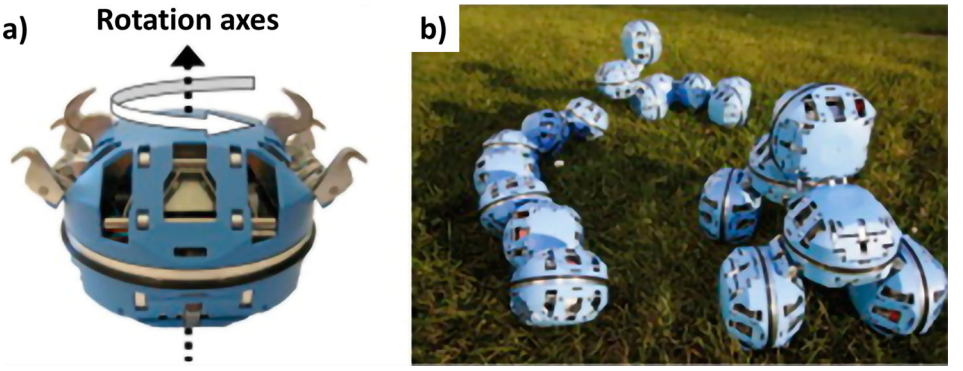


Figure 11. (a) A single ATRON module. (b) Different configurations of ATRON modules: snake-like and quadruped. This image was reproduced from [45] with permission under license number 5839321319589.

The paper [45] by Christensen et al. explored the innovative design of modular and reconfigurable robots. The authors introduced a distributed reinforcement learning strategy that enabled each module within a robotic system to independently optimize its actions based on local information and a shared global reward signal. This morphology-independent approach allowed the robotic system to adapt to various configurations (Figure 11) and recover from module faults effectively. The study's experiments, conducted both in simulation and with physical robots, demonstrated the strategy's capability to facilitate adaptive and efficient locomotion gaits across different robot morphologies, including ATRON and M-TRAN robots. This research underscored the potential of modular and reconfigurable designs in robotics, highlighting their advantages in terms of adaptability, fault tolerance, and ease of implementation.

4. ***Specialized function designs:*** Some QR designs focus on specific functions, such as heavy lifting, climbing, or exploration in hazardous environments. These designs may incorporate unique features like reinforced structures for carrying heavy loads, gripping mechanisms for climbing, or specialized sensors for environmental monitoring. While highly effective in their designated roles, these robots may lack the general versatility found in more broadly purposed designs.

In [48], Y. Li et al. introduced a soft QR driven by inflatable soft legs (Figure 12) based on the pre-charged pneumatic (PCP) actuation principle [54]. Experimental findings revealed that this robotic dog could trot at a maximum speed of 23 cm/s, equivalent to 0.97 body length per second, across various terrains, including slopes, steps, rough, and natural terrains. The robot could support a static load of up to 5.5 kg and carry up to 1.5 kg while trotting. Its soft leg design enabled the robot to endure significant impacts, as evidenced in the paper by its ability to recover and continue trotting after supporting a 73 kg adult (46 times the QR body mass).

5. ***Material-driven designs:*** New material-driven designs are another emerging trend in QRs. Researchers are applying soft robotics principles to build quadrupeds with flexible and durable bodies capable of maneuvering through narrow spaces and adapting to uneven terrain. These concepts use materials such as fluidic actuators or soft polymers to provide an alternative method of locomotion.

The paper [49] by A. A. Stokes et al. investigated the design of a hybrid robotic system that combines rigid and soft subsystems to leverage both advantages. This hybrid system included a wheeled rigid robot (iRobot Create) for rapid locomotion on flat surfaces and a four-legged soft robot (Figure 13) for gripping and retrieval tasks. Made from elastomeric materials, the soft robot demonstrated mechanical compliance, enabling it to perform complex tasks such as gripping fragile objects. The hybrid design enhanced the system's adaptability, allowing it to operate in hazardous environments, such as disaster sites. The study's experimental results demonstrated the hybrid robot's capabilities, highlighting its potential applications in search and rescue, assisted living, and other challenging scenarios.

To conclude, the structural design of QRs is a multifaceted domain, with each approach offering unique advantages and facing distinct challenges. The intended application, operational environment, and specific performance requirements profoundly influence the choice of design.

Control mechanisms in quadruped robotics

At the core of a QR's operational system are its control mechanisms, which determine its stability, flexibility,

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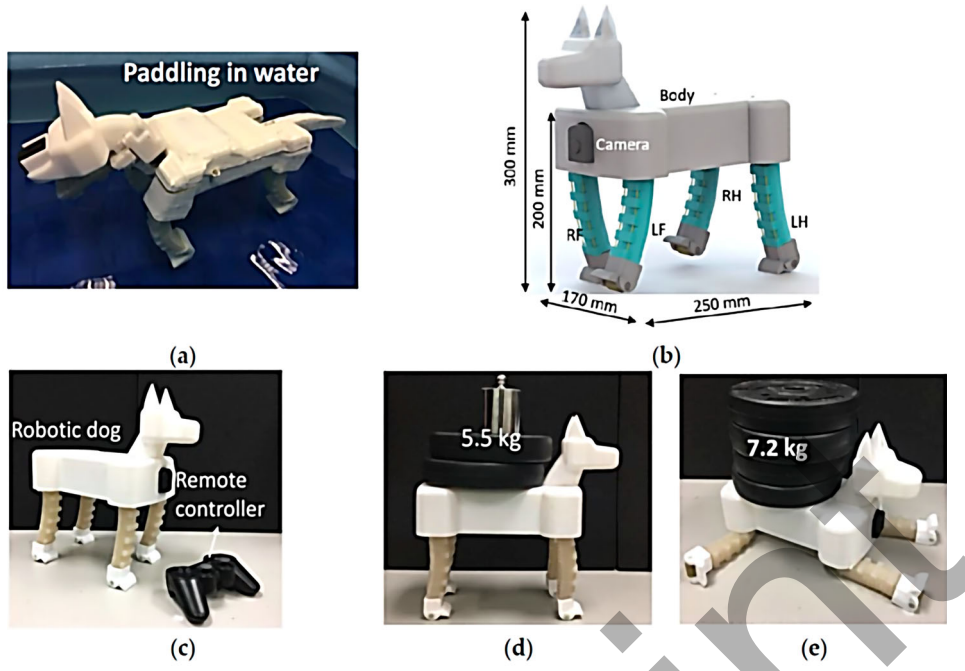


Figure 12. The suggested robotic quadruped in [48]. (a) The soft robotic dog swimming in water. (b) Three-dimensional model of the latest soft robotic canine design. (c) Leg deformation upon landing from a 20 cm free fall. (d) Bearing a static weight of 5.5 kg. (e) Significant leg deformation due to a heavy load. This image was reproduced from [48] (open access).

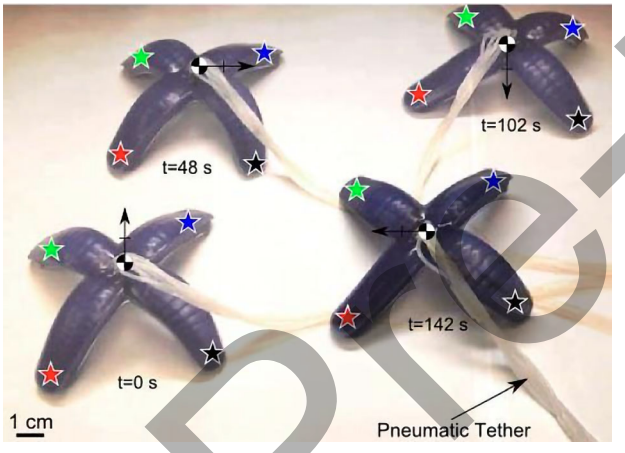


Figure 13. Motion tracking of soft quadruped. This image was reproduced from [49] (open access).

and effectiveness [55]. This section presents a summary of the different control system architectures employed in QR, followed by an examination of both conventional and cutting-edge control strategies. In recent years, the development and implementation of control system architectures in QRs have made significant progress. These control systems can be categorized into centralized and decentralized architectures, each presenting unique advantages and challenges. Centralized control systems operate with a single processing unit, ensuring the synchronization and coordination of limb movements. Despite being easier to manage, they run the risk of creating single points of failure and becoming processing bottlenecks. Conversely, decentralized systems disperse control among several processors, which increases coordination complexity but improves adaptability and resilience, especially in unstructured environments. Further complexity is introduced when examining control strategies, which range from traditional approaches like PID controllers to advanced machine learning and hybrid systems. Traditional control strategies in quadruped robotics often revolve around well-established techniques like Proportional–Integral–Derivative (PID) controllers. Table 3 presents a comparative study of various control mechanisms, highlighting their advantages and disadvantages. It also includes a list of relevant references.

Proportional-integral-derivative (PID) controllers

Robotics widely uses PID control as a feedback mechanism due to its simplicity and effectiveness in maintaining system stability. PID controllers often regulate joint angles and balance in QR, ensuring smooth and well-coordinated robot movements.

The control of a hydraulically driven robotic leg for the 'hyQ' QR was explored by M. Focchi et al. in [77], focusing on two degrees of freedom (q_1 and q_2) at the hip and knee joints. The paper developed experiments to create a full model of the leg and compared various control strategies, including single-input single-output

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Table 3. Comparative study of control mechanisms and list of relevant references.

Control strategy	Strategy	Pros	Cons
Traditional control strategies	PID controllers 	- Simple and well-understood mechanism. - Effective in maintaining system stability. - Reliable for predictable, structured tasks.	- Limited adaptability to dynamic environments. - Struggles with complex, non-linear systems. - Requires precise tuning of parameters.
	Other traditional methods 	- Established techniques with proven track records. - Consistent performance in controlled settings. - Generally easier to implement and debug.	- Lack of flexibility in handling novel situations. - Often requires manual adjustments for different tasks. - May not optimize for efficiency or speed.
Advanced control strategies	Machine learning-based control 	- High adaptability to changing environments. - Ability to learn and improve over time. - Can handle complex, non-linear systems effectively.	- Requires, in general, extensive training data and time. - Potentially unpredictable behavior. - Can be computationally intensive.
	Model predictive control (MPC) 	- Anticipates future states for better planning. - Adaptable to dynamic and unpredictable settings. - Enhances efficiency and precision of movements.	- Computationally demanding. - Requires accurate modeling of robot dynamics and environment. - Challenging to implement in real-time applications.
	Hybrid control systems 	- Balances reliability of traditional methods with adaptability of advanced techniques. - Offers flexibility and learning capabilities. - Suitable for a wide range of applications.	- Complex to design and calibrate. - Requires careful integration of different control paradigms. - Potentially higher development and maintenance costs.

Other related works

PID controllers

Ref	Focus	Key findings	Methods	Robot model
[56]	Development and implementation of a trotting control algorithm for QR	Achieved desired path following within 10 iterations; enhanced trunk attitude with reduced pitch and roll angles to less than ± 3.7 degrees	Hybrid control approach integrating Proportional Integral Derivative and Iterative Learning Control (PID-ILC)	QR (unspecified)
[57]	Design of a stable and intelligent controller for a QR	Fuzzy controller outperformed PID in stability and control efficiency; validated through Lyapunov stability criteria	PID and Fuzzy controllers optimized with ant colony optimization algorithm	QR (unspecified)
[58]	Effects of passive and active joint compliance in quadrupedal locomotion	Enhanced gait performance and stability through optimal passive and active compliance configurations	Tegotae-based control, passive compliance elements	Modular QR (unspecified)

Machine learning (ML)-based control

Ref	Focus	Key findings	Methods	Robot model
[59]	Machine learning strategy for enhancing quadrupedal trot gait speed	Achieved the fastest trot gait for Sony Aibo using policy gradient reinforcement learning, outperforming manually coded and learned gaits	Policy gradient reinforcement learning, parameterized walk, autonomous learning	Sony Aibo
[60]	Learning-based algorithms for agile quadruped locomotion	Developed a hierarchical learning framework for agile locomotion without pre-training, performing effectively across various terrains	Hierarchical learning framework, reinforcement learning, trajectory generator, simulation-based training	QR (unspecified)
[61]	Locomotion control in legged robots using reinforcement learning	Demonstrated exceptional agility, robustness, and adaptability in locomotion; achieved speeds up to 4 m/s with natural gait transitions	Reinforcement learning, reward mechanisms, life-long learning principles	QR (unspecified)
[62]	Controlling the Solo12 QR with deep reinforcement learning	Implemented a robust end-to-end learning-based controller for the Solo12 quadruped using deep reinforcement learning; showed effective transfer to the real robot with energy-efficient control	Deep reinforcement learning, proximal policy optimization (PPO), joint impedance control, domain randomization	Solo12

(*continued*).

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Table 3. Continued.

Machine learning (ML)-based control				
[63]	Reinforcement learning-based stable jump control method for asteroid-exploration QRs	Developed a model-free stable jump control method for asteroid-exploration QRs using multi-agent reinforcement learning (MARL) to handle the unique control challenges posed by irregular gravitational fields on asteroids	Multi-agent reinforcement learning (MARL), jump control, simulated training environment	QR (unspecified)
[64]	Robust quadruped jumping via deep reinforcement learning	Proposed a deep reinforcement learning framework that integrates motor dynamics and power constraints for effective sim-to-real transfer of dynamic jumping motions in QRs	Soft Actor-Critic (SAC) reinforcement learning, trajectory optimization, motor dynamics, power constraints	Unitree A1
[65]	Lifelike agility and play in quadrupedal robots using reinforcement learning and generative pre-trained models	Introduced a hierarchical reinforcement learning framework with deep generative models to produce lifelike agility and strategic behaviors in quadrupedal robots	Deep reinforcement learning, generative pre-trained models, multi-level training, vector-quantized primitive motor control (VQ-PMC)	MAX robot
[66]	Learning robust autonomous navigation and locomotion for wheeled-legged robots	Developed a robust learning framework for autonomous navigation and locomotion in wheeled-legged robots using reinforcement learning and various perception modules	Reinforcement learning, neural network models, hierarchical control, localization techniques	Wheeled-QR (unspecified)
[67]	ANYmal parkour: Learning agile navigation for quadrupedal robots	Achieved advanced locomotion capabilities for quadrupedal robots in challenging environments, using a combination of reinforcement learning and hierarchical control strategies	Hierarchical reinforcement learning, reward shaping, perception modules, custom training environments	ANYmal D
[68]	Differentiable Optimal Control for Retargeting Motions onto Legged Robots	Presented a differentiable optimal control (DOC) framework for retargeting expressive motions from animals or animations onto legged robots of varying proportions and mass distributions	Differentiable optimal control (DOC), model-predictive control (MPC), motion retargeting, continuous optimal control	Multiple quadrupedal robots (varied)
Model predictive control (MPC)				
[69]	Balance problem in hydraulic QRs under lateral disturbances	Developed push recovery control system using MPC; validated stability and adaptability through simulation and experimental trials	Simplified dynamic model (LIP), capture point (CP), zero moment point (ZMP), model predictive control (MPC)	Hydraulic QR (unspecified)
[70]	Whole-body nonlinear model predictive control (NMPC) for rigid body systems in contact interactions	Achieved high execution rates (up to 190 Hz) and significant performance enhancement; validated through hardware experiments on various quadrupeds	Whole-body nonlinear model predictive control (NMPC), comprehensive dynamic system model, explicit contact dynamics	QRs with distinct actuation systems (unspecified)
[71]	Dynamic legged locomotion with aerial phases and rapid leg movements	Combined WBC with MPC for high-speed dynamic locomotion; achieved robustness and adaptability across various environments and gaits, reaching speeds up to 3.7 m/s	Whole-body control (WBC), model predictive control (MPC), reaction force commands, joint torque/position/velocity commands	Mini-Cheetah QR
Hybrid control systems				
[72]	Hybrid systems theory for designing feedback controllers for quadruped locomotion	Introduced virtual constraint controllers for continuous-time quadruped locomotion; demonstrated efficacy via numerical evaluations on Vision 60 robot	Hybrid systems theory, multi-domain hybrid models, virtual constraint controllers, Poincaré return map, iterative optimization	Vision 60 QR
[73]	Model predictive control for robotic systems involving contact	Introduced HiLQR-MPC for handling hybrid dynamical systems; demonstrated superior performance over centroidal methods in simulation and hardware tests	Hybrid iterative linear quadratic regulator (HiLQR), model predictive control (MPC), state jumps, gradient computation strategies	Unitree A1 QR
[74]	Hybrid learning architecture for fuzzy control	Developed a hybrid learning architecture combining reactive behaviors with deliberative reasoning, using FLCs and a two-stage learning scheme	Fuzzy logic controllers (FLCs), Q-learning, genetic algorithm	Sony walking robots

(*continued*).

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Table 3. Continued.

		Hybrid control systems		
[75]	Advancing Locomotion Control of a QR: Harnessing Hybrid Control Algorithms and Digital Twin Technology	Proposed a hybrid dynamic control method that combines nonlinear model predictive control (NMPC) and an adaptive neural network-based fixed-time nonsingular terminal sliding mode control (NTSMC) for precise locomotion control in QRs	Nonlinear model predictive control (NMPC), adaptive neural network, fixed-time nonsingular terminal sliding mode control (NTSMC), digital twin technology	QR (unspecified)
[76]	Hybrid learning mechanisms under a neural control network for various walking speed generation of a QR	Introduced a neural control framework integrating probability-based black-box optimization (PIBB) and supervised learning for generating motor patterns at different walking speeds	Neural control framework, probability-based black-box optimization (PIBB), central pattern generator (CPG), radial basis function (RBF), hypernetwork	Laikago

(SISO), multiple-input multiple-output (MIMO), linear, and nonlinear. It specifically investigated a regular PID controller (linear SISO), an LQR controller (linear MIMO), and a feedback linearization (FL) controller (nonlinear MIMO). Findings showed that LQR performed well at low frequencies but less so at higher ones, while FL showed quick responses in simulations but required adjustments for practical use due to sensitivity to parameter uncertainties. This work highlighted the challenges and considerations in developing effective control algorithms for robotic mobility.

Machine learning-based (ML) control

QRs are increasingly leveraging ML techniques, particularly reinforcement learning (RL), to optimize their control and adaptability to varying terrains and obstacles. RL methods involve training robots through simulations or real-world interactions to autonomously adjust their gaits based on feedback from actions, framed as rewards or penalties. One widely adopted RL technique is Q-learning [78], a value-based approach where the robot learns a policy mapping states to actions by maximizing cumulative rewards over time. Deep Q-Networks (DQN), an extension of Q-learning, integrate deep neural networks to approximate the Q-value function, enabling efficient handling of high-dimensional state spaces, such as those encountered in dynamic and unstructured environments. A study by Ji et al. [79] employed an ML-based approach to optimize gaits for QRs equipped with tendon-driven soft actuators using DQN. Unlike traditional QRs with rigid components, this research highlighted the benefits of soft actuators. However, the non-linear dynamics of soft-legged robots posed significant modeling and control challenges. To address this, the authors combined finite element methods (FEM) and lumped parameter modeling to simulate the robot's behavior while ensuring computational efficiency. The soft actor-critic (SAC) algorithm, a model-free RL approach, was applied to learn optimal walking gaits across various tasks. Preliminary results showed the learned controller enabled effective locomotion, including forward movement and automatic correction of walking direction.

Policy gradient reinforcement learning (PGRL) methods [59,80], such as proximal policy optimization (PPO) [81] and trust region policy optimization (TRPO) [82], have also gained prominence in continuous action spaces, where fine-grained motor control is necessary, such as in legged robots. These methods directly optimize the control policy by adjusting parameters to maximize expected rewards. In contrast, model-based RL techniques predict the outcomes of actions using learned environment models, allowing robots to plan ahead and make informed decisions. Model-free methods, such as deep deterministic policy gradient (DDPG) [83] and SAC, focus on learning control policies without an explicit model, offering more flexibility in environments where dynamics are challenging to model. Among the early works applying PGRL to QRs, Kohl and Stone [59] demonstrated its efficacy in optimizing the gait of the Sony Aibo for maximum forward speed. Their method automated the traditionally labor-intensive process of hand-tuning gaits by iteratively adjusting parameters based on real-world performance. In just three hours of training, their approach achieved a gait faster than any previously known, whether hand-tuned or learned, highlighting the potential of PGRL to significantly enhance the efficiency and adaptability of QR locomotion.

More recently, multi-agent reinforcement learning (MARL) [84,85] has gained traction, especially for scenarios requiring the coordination of multiple robots. In MARL, multiple agents learn either cooperatively or competitively to achieve shared goals or optimize their individual performance. For instance, J. Qi et al. [63] tackled the challenge of planetary rovers navigating irregular, weak gravitational fields of asteroids. Their study proposed legged robots capable of jumping as a

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superior alternative to traditional traction methods. The researchers developed a stable jumping control strategy using MARL, which outperformed single-agent RL in managing complex motion control problems under the same conditions. Task segmentation, tailored reward functions, and simulated gravitational field models supported jump path planning. Simulations validated the effectiveness of this method on asteroid 216 Kleopatra, demonstrating its potential for future space exploration.

The wide variety of RL techniques have enabled QRs to achieve robust locomotion, autonomously adapt to complex environments, and perform tasks with minimal human intervention.

Model predictive control (MPC)

Another advanced approach is MPC, in which the control system predicts the robot's future states based on a model of its dynamics and environment [55]. This predictive capability allows the robot to plan its movements more effectively, especially in dynamic and unpredictable settings [55].

A comprehensive control methodology designed for enhancing the cooperative locomotion of two holonomically constrained QRs through real-time trajectory planning and control was presented by J. Kim et al. in [86]. The main idea behind this method was to create a network of simplified models that were all based on the dynamics of a single rigid body (SRB). These models were designed to work together during the trajectory planning phase. To find the best way to control the interconnected SRB dynamics, they used a two-tier control architecture that combined centralized and distributed Model Predictive Control (MPC) algorithms. They particularly used the distributed MPC due to its reliance on local quadratic programs, which synchronized their optimal outcomes after a one-step communication delay. In addition, the authors carefully developed distributed nonlinear controllers at the inferential level of the control hierarchy to make sure that the full-order dynamics matched the paths set by the MPCs. Extensive numerical simulations and experimental trials corroborated the efficacy of this layered control approach, demonstrating its ability to maintain robust and cooperative locomotion for two A1 QRs, each carrying distinct payloads, across varied terrains and amidst disturbances.

Hybrid control systems

Many modern QR systems combine traditional methods like PID with advanced techniques like machine learning. This approach balances the reliability of traditional control mechanisms with the adaptability and learning capabilities of modern algorithms.

Y. Zhang et al. [76] presented a neural control framework that enables legged robots to efficiently adapt their motor patterns for varying walking speeds. The framework combined supervised learning and black-box optimization to create a neural CPG-RBF-hypercontrol network. The authors did this by connecting a central pattern generator (CPG), a radial basis function (RBF)-based premotor network, and a hypernetwork. Initially, the CPG-driven RBF network learned different speed policies through an incremental learning strategy. The hypernetwork then mapped these policies to the QR's walking speed, generating optimal motor joint patterns. This innovative approach enabled a QR to walk stably and robustly at varied speeds without sensory feedback, demonstrating significant transferability from simulation to real-world application. Testing on untrained CPG frequencies further confirmed the controller's generalization ability, highlighting its efficiency and versatility in QR locomotion.

Finaly, the control mechanisms in quadruped robotics are as diverse as their mechanical designs. The QR's application often shapes the choice of control architecture and strategy, which is crucial. As the field advances, we expect the integration of advanced control systems to further enhance the capabilities and autonomy of QR, empowering them to perform more complex tasks in diverse environments.

Sensing and perception

Sensing and perception capabilities are fundamental to the functionality of QR, enabling them to interact with and navigate through their environment intelligently [87]. This section explores the role of various sensors, their integration, and the advances in sensor fusion techniques that enhance the perception abilities of these robots.

Sensors in QR serve as the primary interface between the robot and its environment [27,88]. They gather crucial data that inform the robot's decisions and movements. The primary functions include:

- *Obstacle Detection and Avoidance:* Proximity sensors, LiDAR (Light Detection and Ranging), and ultrasonic sensors help in detecting obstacles, allowing the robot to navigate safely without collisions.
- *Terrain Analysis:* To adapt to different terrains, QR use a combination of sensors to assess ground conditions, such as incline, texture, and stability. This information is crucial for adjusting gait and balance.
- *Localization and Mapping:* GPS and inertial measurement units (IMUs) are used for outdoor localization, while indoor environments might rely on visual

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SLAM (Simultaneous Localization and Mapping) techniques.

T. Miki et al. in [6], studied the potential of legged robots to autonomously navigate remote and hazardous environments, enhancing exploration capabilities. Researchers identified exteroceptive perception as the key to achieving rapid and energy-efficient locomotion, as it enables terrain recognition and gait adaptation prior to contact, thereby maintaining speed and stability. Even though it's important, it has been reported that firmly implementing exteroceptive perception in four-legged locomotion has been a big problem because of things like snow, plants, and water, as well as problems with depth perception in bad lighting or when sensors are blocked. As a result, most current legged locomotion solutions rely on proprioception, which restricts speed because robots must physically contact terrain to adjust their gait. The paper described a strong solution that combines exteroceptive and proprioceptive perception. The system achieved this by utilizing an attention-based recurrent encoder, which undergoes comprehensive training to integrate various forms of perception (Figure 14).

Sensor systems are categorized based on their types in Table 4, their key functions in quadruped robotics are outlined, and the recent technological advances enhancing their capabilities are highlighted. The table also includes additional works, detailing their focus, key findings, methodologies, and specific robot models used.

The last five works mentioned in the Table 4 provided a preview of recent advancements in sensor fusion techniques. Multi-modal fusion, which integrated data from vision and tactile sensors, enhanced object recognition and manipulation capabilities. Advances in computing power and algorithms facilitated real-time processing of sensor data, which was crucial for operation in dynamic and unpredictable environments. The application of machine learning to interpret sensor data, learn from experiences, and make predictive judgments significantly improved the adaptability of QRs. Furthermore, more advanced fusion techniques can build strong QR systems that could fix sensor problems or compensate for their errors. Additionally, the integration of IoT technologies enabled continuous communication between QRs and external devices, allowing for remote monitoring, control, and data exchange, which further enhanced their operational efficiency and responsiveness [95].

Challenges and limitations

One of the major hurdles lies in the complexity of design; creating QR involves a delicate interplay between mechanical and electronic components, working towards a balance that ensures stability, mobility, and functionality. QRs must adapt to diverse terrains and execute multiple tasks, thereby pushing the boundaries of current design methodologies, as demonstrated in [96]. In parallel, control system challenges emerge as another significant barrier. These systems must navigate the unpredictable nature of real life by processing vast amounts of sensor data in real time and making precise decisions through machine learning techniques such as reinforcement learning (RL), as employed in [62,97,98]. The quest for autonomous adaptation in these robots, without direct human guidance, marks an important area of ongoing research.

Energy efficiency is another notable challenge in the development of mobile platforms, such as QRs. The design of QR for remote operations presents a significant challenge in minimizing power consumption and maximizing operation duration, primarily because of the limitations imposed by existing battery technologies. This issue directly impacts their practical usability and overall performance capabilities. In [99], researchers developed an appropriate kinematic model of the quadruped and its energy consumption, aiming to use RL to optimize its policies for enhanced performance. This ties into the robustness and durability of the robotic platform, which is why it is critical for improving how these robots operate effectively in varied and harsh environments. Researchers in [100–102] underscored the necessity of robust unmanned robots, particularly quadrupeds, for asset inspection in the energy sector. Mechanical components are prone to wear and tear, whereas electronic systems face threats from environmental factors such as water, dust, extreme temperatures, and radiation, making durability a central design challenge.

The development and deployment of QRs pose significant challenges, emphasizing the complex tasks of converting from conceptual designs to operational real-world machines. The main obstacles include achieving intuitive design and control systems, ensuring energy efficiency and durability [103], and integrating advanced sensor technologies. These challenges encompass technical issues, scalability, cost, and the broader implications of autonomy and human-robot interaction. Addressing these challenges is crucial for advancing QRs and ensuring their successful integration into everyday human activities. This pursuit of innovation promises a future where QRs enhance human productivity, safety, and exploration.

One other major obstacle is the limitations of sensors. While sensor technology has advanced significantly, problems with accuracy, range, and reliability still exist. Examples of such limitations are discussed in

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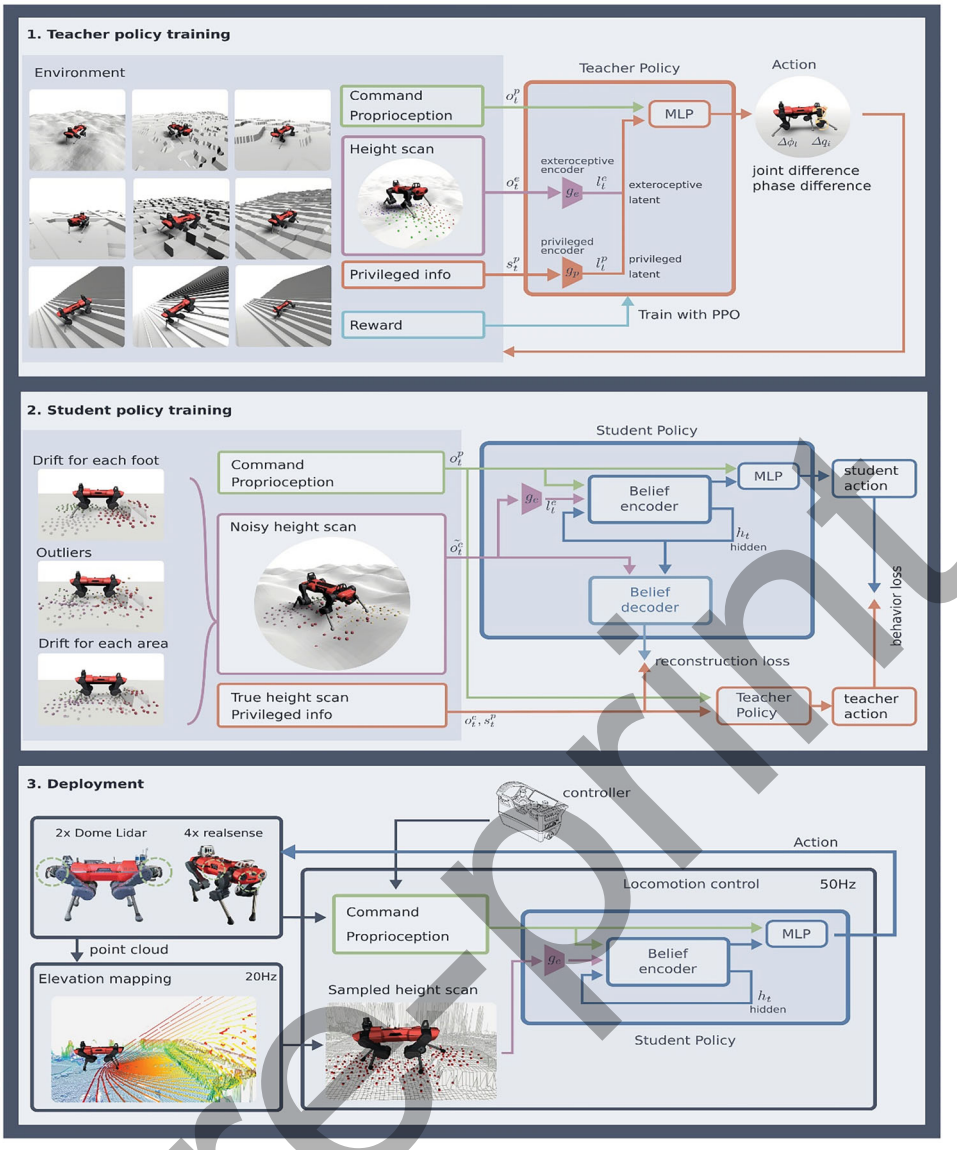


Figure 14. Schematic showing the training process and deployment using sensing system implemented in [6]. This image was reproduced from [6] (open access).

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[104], which pointed out the benefits and drawbacks of employing sensors in environments that minimize their efficacy and methods of improving performance. Other significant limitations include scalability and cost concerns. Scaling the use of QRs while maintaining their affordability and functional integrity is challenging due to the high production costs associated with their inherent level of sophistication and complexity in design and manufacturing.

Improving autonomy and AI capabilities in QRs continues to be a significant challenge [105]. Although there have been advancements in AI and machine learning, these robots frequently rely on human intervention or supervision, particularly when confronted with highly complex tasks. The need for improved human-robot interaction, especially in situations where human involvement is required, further complicates this challenge. To ensure that QR can safely and intuitively interact with humans, especially in public or home environments, it is imperative to make advancements in their capacity to comprehend and anticipate human behaviors, as well as uphold safety protocols.

The ethical and social implications of deploying quadruped robotics are gaining more attention [106,107]. As these technologies gain wider acceptance, it is critical that we handle concerns about privacy, potential job displacement, and misuse with caution and care. In order to ensure the responsible development and integration of QR technology into society, it is imperative that we also address the ethical and social considerations associated with it. This involves carefully considering the potential impacts on human lives and societal

Table 4. Overview of sensor systems in quadruped robotics: types, functions, technological advances and additional research works.

Sensor type	Primary functions	Recent advances
Vision sensors (e.g. RGB Cameras, depth cameras)	- Object recognition and tracking. - Navigation and mapping. - Interaction with environment and entities.	- Enhancement in 3D imaging and depth perception. - Integration with AI for better object identification. - Improved low-light performance and resolution.
Tactile sensors (e.g. Pressure sensors, skin-like sensors)	- Surface texture and property detection. - Delicate object manipulation. - Interaction feedback for safe contact.	- Development of more sensitive and resilient materials. - Integration with haptic feedback for realistic sensation. - Increased spatial resolution for detailed surface mapping.
Proprioceptive sensors (e.g. Joint encoders, IMUs)	- Internal status monitoring (limb position, orientation). - Balance and posture control. - Coordination of movement.	- Enhanced precision and response rate. - Miniaturization for integration without hindering mobility. - Improved calibration techniques for more accurate readings.
LiDAR and ultrasonic sensors	- Obstacle detection and avoidance. - Terrain mapping. - Distance and speed measurement.	- Increased range and accuracy. - Reduction in size and energy consumption. - Fusion with other sensor data for comprehensive environment modeling.
GPS and IMUs (for outdoor robots)	- Localization and wayfinding. - Route planning and navigation. - Speed and direction monitoring.	- Higher accuracy and reliability in varied environments. - Integration with cellular and satellite technologies. - Real-time data processing for dynamic path adjustments.

Other related works

Ref	Focus	Key Findings	Methods	Robot Model
[89]	Adaptive sensor foot for bipedal and quadrupedal robots	Enhanced traction and adaptability on varied terrains; supported transitions between bipedal and quadrupedal locomotion	6-axis force/torque sensor, pressure sensor array, digital angular encoders, biologically inspired design	Bipedal and quadrupedal robot prototype
[90]	Positioning of QR using tightly coupled LiDAR, vision, and IMU data	Achieved high precision and strong robustness in positioning for both indoor and outdoor environments	Tightly coupled LiDAR Vision Inertial Odometer (LVIO), nonlinear optimization, loopback detection	QR (unspecified)
[91]	Multisensor fusion for legged robot state estimation	Achieved high accuracy in state estimation across various terrains, demonstrated on ANYmal	Factor graphs integrating visual, inertial, lidar, and leg odometry sensors	ANYmal B300, ANYmal C100
[92]	Localization and situational awareness methods for biped and QRs	Comprehensive review of sensors for state estimation; developed an estimator for Atlas and adapted for HyQ2	Kinematics, pressure sensors, accelerometers, gyroscopes, vision, LIDAR	Atlas humanoid robot, HyQ2 QR
[28]	Application of biological concepts for quadruped locomotion	Achieved adaptive walking on irregular terrain using virtual spring-damper systems and CPG-based control	PD-controller, central pattern generator (CPG), multi-sensor integration, vision	QR (unspecified)
[26]	Comprehensive state estimator for quadrupedal robots	Developed a low-level state estimator for attitude, odometry, ground reaction forces, and contact detection; validated with HyQ robot	Nonlinear observer, encoders, force sensors, torque sensors, inertial data integration	HyQ QR
[93]	Ground substrate classification for adaptive locomotion	Achieved 99.83% classification accuracy during training and 98.33% in real-time testing using a multi-sensor fusion method	Back Propagation Neural Network (BPNN), Center of Gravity (COG) adjustment, 6-axis motion sensor	Biodog QR
[94]	Obstacle detection and motion distortion correction using multi-sensor fusion	Improved obstacle detection and mapping accuracy by integrating LiDAR, camera, and odometer data; addressed LiDAR motion distortion	Point-map fusion (PMF) algorithm, motion distortion removal (MDR) method	QR dog (unspecified)

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structures. The work done in [108] shows how important it is for policymakers, ethicists, technologists, and other relevant people to understand how important it is to work together to set up the framework needed to handle the responsibilities that come with the fourfold rise in robotics and how they affect society as a whole.

An overview of the multiple facets where challenges and limitations are present in the field of quadruped robotics is given in Table 5.

Future directions and potential applications

Recent trends imply a shift toward increasing robot autonomy, allowing them to perform complex tasks without human intervention. Collaborative robots (COBOTS) [109,110] that can safely and effectively work with humans or other robots are also gaining popularity. Quadruped robotics will likely emphasize autonomy. This change will enable robots to perform more complex tasks independently, such as precise manipulation and autonomous decision-making in unpredictable environments like construction sites. These features enhance QR's utility in catastrophic scenarios such as disaster relief, radiation detection, and high-risk work, necessitating swift and autonomous actions. COBOTS represent a new robotics direction that integrates seamlessly into human teams [111,112]. This collaborative approach boosts human efforts and improves shared workspace safety and efficiency. QRs could also help deliver medications and care for patients in contagious

Table 5. Overview of the challenges and limitations in the field of quadruped.

Area of focus	Challenges	Limitations
Design	- Balancing stability, mobility, and functionality. - Adapting to various terrains and tasks.	- Wear and tear in harsh environments. - Integration of complex mechanical and electronic systems.
Control systems	- Handling unpredictable environments. - Real-time processing of large sensor data volumes. - Autonomous learning and adaptation.	- Limited decision-making capabilities in complex scenarios. - Dependence on human oversight or intervention.
Energy efficiency	- Managing high power consumption for prolonged use. - Balancing performance with battery life.	- Limitations in current battery technology affecting operational time.
Robustness and durability	- Ensuring long-term operation in diverse conditions. - Protecting against environmental elements.	- Challenges in making durable yet lightweight components.
Sensor technology	- Enhancing range, accuracy, and reliability. - Adapting sensors to function under varied conditions.	- Current sensors may have limitations in range and accuracy.
Scalability and cost	- Reducing manufacturing and development costs. - Making quadruped robotics accessible and scalable.	- High costs associated with sophisticated designs and technologies.
Autonomy and intelligence	- Achieving high-level autonomy. - Integrating advanced AI and machine learning effectively.	- Current AI limitations in fully autonomous operations.
Human-robot interaction	- Ensuring safe and intuitive interactions. - Predicting and understanding human behavior.	- Challenges in developing robots that safely coexist with humans in various settings.
Ethical and social considerations	- Addressing privacy, job displacement, and misuse potential. - Developing regulations and standards.	- Ethical concerns with widespread deployment and integration into society.

disease wards, reducing human risk. Studies such as [113] on the efficacy of robotics to assist caregivers are also beneficial to healthcare. These advancements will allow QRs to perform previously unattainable tasks in the military, environmental monitoring, and space exploration. As the autonomy and application range of QR expand, so does the importance of human robot interface (HRI) [114–116]. Future QR will likely possess advanced sensory and processing capabilities to understand and respond to human commands more effectively, recognize emotional cues, and adapt their behaviors accordingly. This evolution will facilitate safer and more intuitive interactions in shared spaces, enhancing the collaboration between humans and robots. The development of such sophisticated HRI mechanisms is crucial for the successful integration of QR into daily human activities and work processes. In [117], the task of construction inspection and monitoring being aided by robotic platforms was discussed focusing on the lack of existing knowledge regarding human robot interaction. Also, in the study performed in [107], the initial human-robot encounter plays a role in how humans interact and develop trust for a robotic platform while understanding the immediate used of the technology in their daily live. Quadruped robotics raises many ethical and regulatory issues. To ensure that these technologies improve society without violating individual rights, privacy, security, and employment issues must be carefully considered. Comprehensive standards and regulations are necessary for the safe and responsible integration of QR into various sectors [118–121].

Conclusion

This systematic review examines the design, control dynamics, and advancements in sensing and perception technologies that shape the current and future development of QRs. Our exploration showed the symbiosis between bio-inspired design principles and complex mechanical configurations that aim to mimic nature. AI, ML and MPC innovated autonomous and adaptive locomotion, allowing robots to navigate unexplored environments with a high degree of agility. The integration of advanced sensing systems, from tactile sensors to LiDAR, has opened new avenues for environmental interaction, facilitating a deeper understanding and engagement with the world. Sensor fusion plays a vital role in enhancing perception capabilities, coupled with cutting-edge control algorithms, paves the way for QRs to accurately undertake tasks across various domains, from disaster recovery to planetary exploration. The quest for energy efficiency and seamless human-robot interaction remains at the heart of research efforts.

In conclusion, we hope that this systematic review contributes to the understanding of the significant advancements in robotics towards the realization of robots that transcend the boundaries of what is currently conceivable.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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